Linear Number Patterns – Extra Practice

These questions are extra practice in the same style as Questions 1 and 2 of the *Linear number patterns* worksheet. For each question: (a) draw the next pattern, (b) complete the table, (c) find a formula, then (d)–(e) use your formula to answer questions about much larger patterns. Questions 1–3 are *linear* (add a constant amount each step); Question 4 is *geometric* (multiply by a constant amount each step).

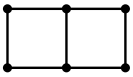
Question 1 – Matchstick Squares

This pattern shows the number of matchsticks needed to make a row of joined squares.

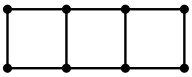
1:



2:



3:



(a) Draw the next pattern (4 squares in a row).

Draw your pattern here.

(b) Copy and complete the table.

S		1	2	3	4	5	12
M							

(c) Construct a formula for the number of matches (M) needed for any number of squares (S).

(d) If a row had 40 squares, how many matches would be needed?

(e) If a row had 75 squares, how many matches would be needed?

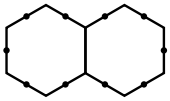
Question 2 – Hexagonal Picnic Tables

A campsite joins hexagonal picnic tables end to end in a row, as shown below (each dot is a seat). When two tables are pushed together, the shared side can no longer seat anyone.

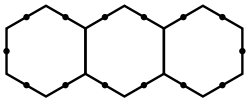
1:



2:



3:



(a) Draw the next pattern (4 hexagonal tables joined in a row).

Draw your pattern here.

(b) Copy and complete the table below.

T		1	2	3	4	5	25
C							

(c) Construct a formula for the number of seats (C) needed for any number of tables (T).

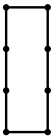
(d) If the campsite joined 40 tables in a row, how many seats would there be?

(e) If the campsite needed to seat 70 people in a single row of joined tables, how many tables would be needed?

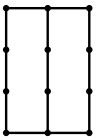
Question 3 – Matchstick Towers

This pattern shows the number of matchsticks needed to make a row of joined towers (each tower is 1 matchstick wide and 3 matchsticks tall). Neighbouring towers share a whole wall – 3 matchsticks' worth of overlap.

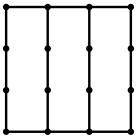
1:



2:



3:



(a) Draw the next pattern (4 towers in a row).

Draw your pattern here.

(b) Copy and complete the table.

T		1	2	3	4	5	20
M							

(c) Construct a formula for the number of matches (M) needed for any number of towers (T).

(d) If a row had 30 towers, how many matches would be needed?

(e) If a row had 60 towers, how many matches would be needed?

Question 4 – Paper Cutting (Geometric)

A square sheet of paper is cut with straight vertical cuts, each one cutting every piece made so far exactly in half. This pattern is *geometric*, not linear: the number of pieces is *multiplied* by the same amount at each stage.

1:



2:



3:



(a) Draw the next pattern (stage 4).

Draw your pattern here.

(b) Copy and complete the table below.

n	1	2	3	4	5	7
P						

(c) Construct a formula for the number of pieces (P) at stage n. (Hint: write a few terms as powers of the ratio you found above.)

(d) How many pieces are there at stage 8?

(e) How many pieces are there at stage 10?

Section 5 – Sequences in Words

Word problem 1

A candle is 24 cm tall when it is lit. Every hour it burns down, it loses 4 cm of height.

(a) Copy and complete the table below, using the words above.

h		1	2	3	4	5
H						

(b) Construct a formula for the height (H) after h hours.

(c) After how many hours does the candle burn out completely (height $= 0$)?

(d) Your formula gives a height for $h = 8$. Work it out, then explain why this answer doesn't make sense for the real candle.

Word problem 2

A chess club has 15 members at the start of the year. Every week, 4 new people join the club (and nobody leaves).

(a) Copy and complete the table below, using the words above.

w		1	2	3	4	5	25
M							

(b) Construct a formula for the number of members (M) after w weeks.

(c) How many members would there be after 10 weeks?

(d) After how many weeks would the club have 99 members?

Section 6 – Just the Numbers

Pattern A

n	1	2	3	4	5
y	5	9	13	17	21

(a) Find a formula for y in terms of n.

(b) Describe a real-life situation this sequence could represent.

(c) Use your formula to find the 15th term.

Pattern B

n	1	2	3	4	5
y	50	45	40	35	30

(a) Find a formula for y in terms of n.

(b) Describe a real-life situation this sequence could represent.

(c) Use your formula to find the 10th term.

Pattern C

n	1	2	3	4	5
y	3	6	12	24	48

(a) Find a formula for y in terms of n.

(b) Describe a real-life situation this sequence could represent.

(c) Use your formula to find the 8th term.

Pattern D

n	1	2	3	4	5
y	8	8	8	8	8

(a) Find a formula for y in terms of n.

(b) Describe a real-life situation this sequence could represent.

(c) Use your formula to find the 20th term.